

Final Project-Telco Customer Churn

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```
data <- read.csv("C:/Users/brumb/OneDrive/Documents/R/WA_Fn-UseC_-Telco-Customer-Churn.csv")
```

```
str(data)
```

```
## 'data.frame':    7043 obs. of  21 variables:
## $ customerID      : chr  "7590-VHVEG" "5575-GNVDE" "3668-QPYBK" "7795-CFOCW" ...
## $ gender          : chr  "Female" "Male" "Male" "Male" ...
## $ SeniorCitizen   : int   0 0 0 0 0 0 0 0 0 0 ...
## $ Partner         : chr  "Yes" "No" "No" "No" ...
## $ Dependents      : chr  "No" "No" "No" "No" ...
## $ tenure          : int   1 34 2 45 2 8 22 10 28 62 ...
## $ PhoneService    : chr  "No" "Yes" "Yes" "No" ...
## $ MultipleLines    : chr  "No phone service" "No" "No" "No phone service" ...
## $ InternetService : chr  "DSL" "DSL" "DSL" "DSL" ...
## $ OnlineSecurity   : chr  "No" "Yes" "Yes" "Yes" ...
## $ OnlineBackup     : chr  "Yes" "No" "Yes" "No" ...
## $ DeviceProtection: chr  "No" "Yes" "No" "Yes" ...
## $ TechSupport     : chr  "No" "No" "No" "Yes" ...
## $ StreamingTV      : chr  "No" "No" "No" "No" ...
## $ StreamingMovies  : chr  "No" "No" "No" "No" ...
## $ Contract        : chr  "Month-to-month" "One year" "Month-to-month" "One year" ...
## $ PaperlessBilling: chr  "Yes" "No" "Yes" "No" ...
## $ PaymentMethod    : chr  "Electronic check" "Mailed check" "Mailed check" "Bank transfer (automatic)" ...
## $ MonthlyCharges   : num   29.9 57 53.9 42.3 70.7 ...
## $ TotalCharges     : num   29.9 1889.5 108.2 1840.8 151.7 ...
## $ Churn            : chr  "No" "No" "Yes" "No" ...
```

```
summary(data)
```

```
##   customerID      gender      SeniorCitizen      Partner
## Length:7043      Length:7043      Min.   :0.0000      Length:7043
## Class :character  Class :character  1st Qu.:0.0000      Class :character
## Mode  :character  Mode  :character  Median :0.0000      Mode  :character
##                                     Mean   :0.1621
##                                     3rd Qu.:0.0000
##                                     Max.   :1.0000
##
##   Dependents      tenure      PhoneService      MultipleLines
## Length:7043      Min.    : 0.00      Length:7043      Length:7043
## Class :character  1st Qu.: 9.00      Class :character  Class :character
```

```
## Mode :character Median :29.00 Mode :character Mode :character
## Mean :32.37
## 3rd Qu.:55.00
## Max. :72.00
##
## InternetService OnlineSecurity OnlineBackup DeviceProtection
## Length:7043 Length:7043 Length:7043 Length:7043
## Class :character Class :character Class :character Class :character
## Mode :character Mode :character Mode :character Mode :character
##
##
##
## TechSupport StreamingTV StreamingMovies Contract
## Length:7043 Length:7043 Length:7043 Length:7043
## Class :character Class :character Class :character Class :character
## Mode :character Mode :character Mode :character Mode :character
##
##
##
## PaperlessBilling PaymentMethod MonthlyCharges TotalCharges
## Length:7043 Length:7043 Min. : 18.25 Min. : 18.8
## Class :character Class :character 1st Qu.: 35.50 1st Qu.: 401.4
## Mode :character Mode :character Median : 70.35 Median :1397.5
## Mean : 64.76 Mean :2283.3
## 3rd Qu.: 89.85 3rd Qu.:3794.7
## Max. :118.75 Max. :8684.8
## NA's :11
## Churn
## Length:7043
## Class :character
## Mode :character
##
##
##
```

Data Preparation

```
data <- data %>%
  mutate(
    TotalCharges = as.numeric(ifelse(TotalCharges == "", NA, TotalCharges)),
    Churn = as.factor(Churn)
  ) %>%
  na.omit()

data <- na.omit(data)

data$Churn <- as.factor(data$Churn)
```

Train-Test Split

```

set.seed(123)

trainIndex <- createDataPartition(data$Churn, p = 0.7, list = FALSE)
train <- data[trainIndex, ]
test <- data[-trainIndex, ]

train$customerID <- NULL
test$customerID <- NULL

```

Model Development

```

model <- glm(Churn ~ tenure + MonthlyCharges + Contract,
             data = train,
             family = "binomial")

summary(model)

```

```

##
## Call:
## glm(formula = Churn ~ tenure + MonthlyCharges + Contract, family = "binomial",
##      data = train)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -1.575454   0.106717 -14.763  <2e-16 ***
## tenure        -0.034822   0.002426 -14.352  <2e-16 ***
## MonthlyCharges  0.028102   0.001599  17.575  <2e-16 ***
## ContractOne year -1.168247   0.122339  -9.549  <2e-16 ***
## ContractTwo year -1.953762   0.191984 -10.177  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 5702.8  on 4923  degrees of freedom
## Residual deviance: 4310.2  on 4919  degrees of freedom
## AIC: 4320.2
##
## Number of Fisher Scoring iterations: 6

```

Model Evaluation

```

probabilities <- predict(model, test, type = "response")

predictions <- ifelse(probabilities > 0.5, "Yes", "No")
predictions <- as.factor(predictions)

confusionMatrix(predictions, test$Churn)

```

```
## Confusion Matrix and Statistics
```

```
##
##           Reference
## Prediction   No   Yes
##           No 1386 284
##           Yes 162 276
##
##           Accuracy : 0.7884
##           95% CI : (0.7704, 0.8057)
##           No Information Rate : 0.7343
##           P-Value [Acc > NIR] : 5.067e-09
##
##           Kappa : 0.4172
##
## Mcnemar's Test P-Value : 1.007e-08
##
##           Sensitivity : 0.8953
##           Specificity : 0.4929
##           Pos Pred Value : 0.8299
##           Neg Pred Value : 0.6301
##           Prevalence : 0.7343
##           Detection Rate : 0.6575
##           Detection Prevalence : 0.7922
##           Balanced Accuracy : 0.6941
##
##           'Positive' Class : No
##
```

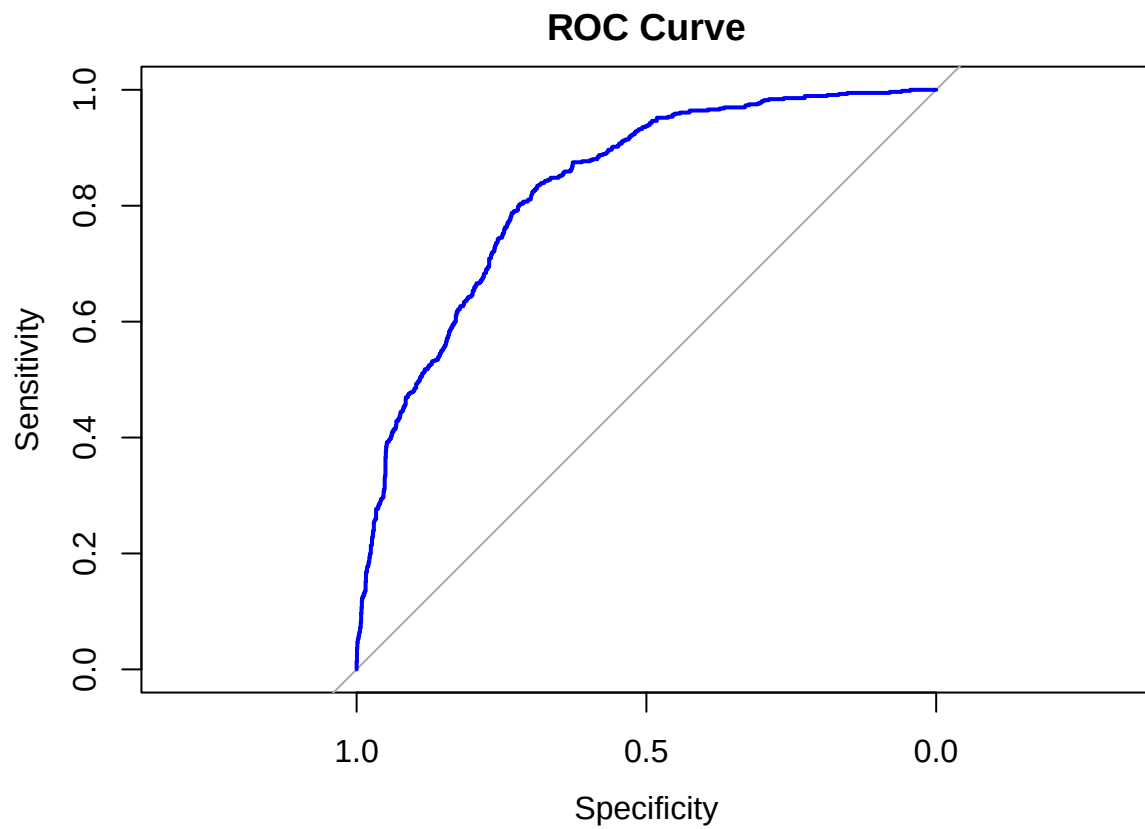
Roc Curve

```
roc_obj <- roc(test$Churn, probabilities)
```

```
## Setting levels: control = No, case = Yes
```

```
## Setting direction: controls < cases
```

```
plot(roc_obj, col = "blue", main = "ROC Curve")
```

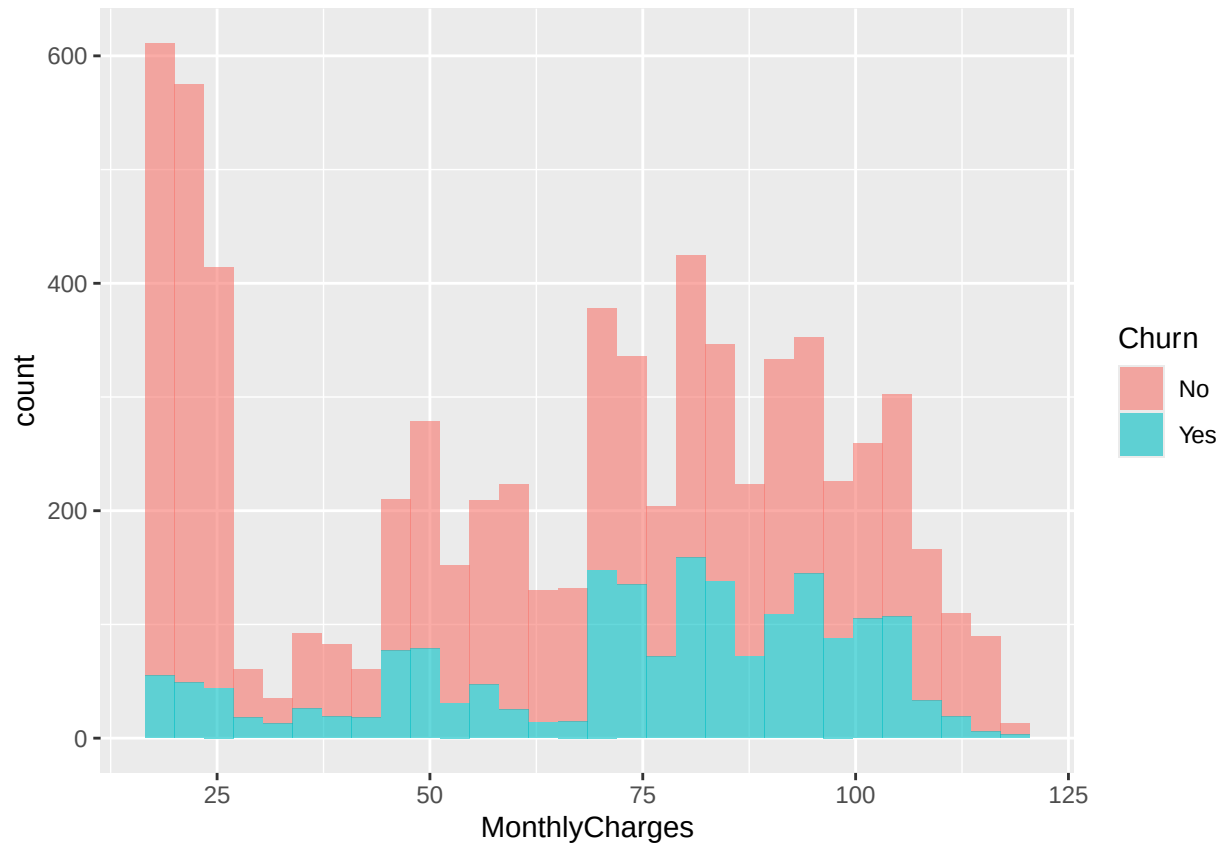


```
auc(roc_obj)
```

```
## Area under the curve: 0.8288
```

History of Monthly Changes

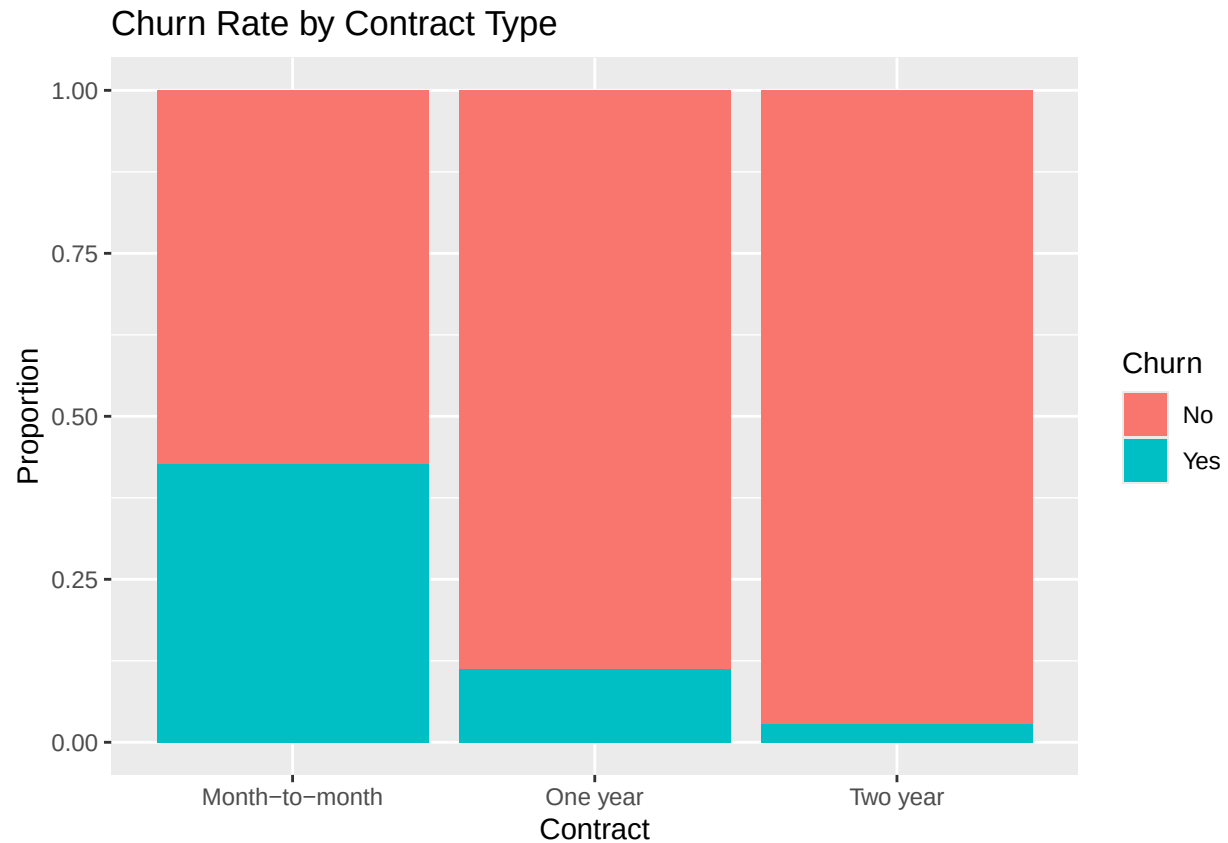
```
ggplot(data, aes(x = MonthlyCharges, fill = Churn)) +  
  geom_histogram(bins = 30, alpha = 0.6)
```



Customers with higher monthly charges are more likely to churn, as seen by the greater concentration of churned customers at higher price levels.

Contact Type vs Churn

```
ggplot(data, aes(x = Contract, fill = Churn)) +  
  geom_bar(position = "fill") +  
  ylab("Proportion") +  
  ggtitle("Churn Rate by Contract Type")
```



Customers on month-to-month contracts have significantly higher churn rates compared to those on long-term contracts.

Odds Ratios

```
exp(coef(model))
```

```
##      (Intercept)          tenure  MonthlyCharges ContractOne year
##      0.2069135         0.9657769         1.0285005         0.3109114
## ContractTwo year
##      0.1417398
```

Higher monthly charges increase churn risk Longer tenure reduces churn Long-term contracts significantly reduce churn